

Labfront Evaluation Report

Exported Data

The output of the data comes in the form of a zip file with several folders pertaining to different measurements, each which contains CSV files with the measurements as specific times and other metadata like the Participant ID, timestamps, Project IDs, Garmin Device IDs and the project title. The major categories are listed below, with those most relevant to this project shown in bold.

Raw Physiological Measurements

- **Beat-to-Beat Intervals**
- **Enhanced Beat-to-Beat Intervals**
- **Heart Rate**

HRV

- **HRV Values**
- **HRV Summary**

Sleep Metrics

- Sleep Summary
- Sleep Stages
- Sleep Respiration
- Sleep Nap Data

Stress Metrics

- Garmin Stress Score
- Device Stress Measurements

Activity

- Activity Details (Summary)
- Activity Summary
- Move IQ Summary
- Daily Summary
- Daily Heart Rate
- Epoch Data

Misc

- Respiration Rate
- Skin Temperature

While Labfronts display methods provide a great deal of information, much of it is not relevant to assessing emotional volatility OR has the black box issue of not allowing us to see what is actually

going into determining score, which introduces some uncertainty and prevents us from fully understanding how those metrics are calculated.

Beat to Beat versus Summaries for Arrival Frequency

With HRV, the most relevant example of this, we would have two options for this, one being to derive our own HRV from the BBI data or using the HRV data Labfront provides. The Labfront data is divided into two: one being HRV Values which are recorded every 5 minutes to get an average rate, and the other being the HRV summary which provides one record per night and additional data points like highest HRV rate (lastNight5MinHigh).

Beat to Beat Intervals (BBI) (garmin-device-enhanced-bbi)

- Raw heartbeat interval data (milliseconds)
- Can compute any HRV metric
- Most flexible option
- Requires additional preprocessing and can be computationally expensive/slow

HRV Values (garmin-connect-hrv-values)

- Garmin-computed HRV measurements (~every 5 minutes)
- Minimal preprocessing required
- Easy to align with EMA prompts
- **Best choice for the current decision engine**

HRV Summary (garmin-connect-hrv-summary)

- Nightly HRV averages and summary statistics
- Reflects overall recovery and chronic physiological state
- Good for baseline monitoring
- Not suitable for real-time intervention decisions

Nightly summaries are unlikely to be useful and not useful for what our project is intending to do which is to provide live intervention. As of now, the HRV values being output every 5 minutes is the most relevant but if that is still too large of an interval, the next option would be manual HRV calculations derived from the enhanced-bbi file, which would allow for greater frequency but could lead to extra computational expense.

Formatting

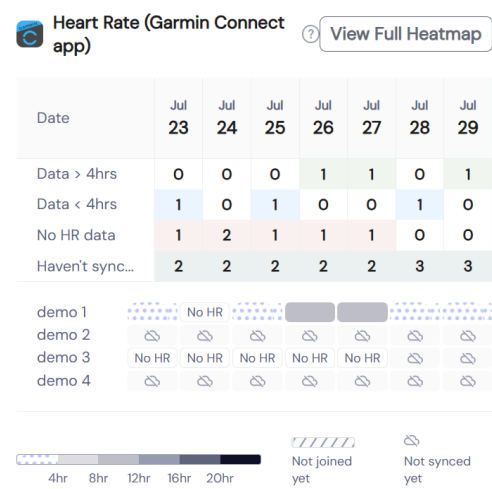
The export format is largely outlined in the Exported Data section above. User data comes as one zip file with each folder corresponding to a different measurement with csv files in each folder.

CSVs are broken down by time but reflect the same data and I'm not sure how Labfront determines when a new CSV file is created. . Having them continuously update in the same CSV would improve the simplicity of the data processing. In the future finding some method to access the data quicker through an API or not having to use CSV would definitely improve efficiency with large datasets.

Non-Wear Detection

The exported data did not contain any explicit wear status field,meaning non-wear cannot be directly identified from the CSV files alone. Despite this, there was one relevant indicator found in the garmin-device-enhanced-bbi export, which includes a confidence field with binary values (0 and 1). When I looked closer I found that BBI measurements continued even when the confidence value changed from 1 to 0, suggesting that this field reflects the quality or reliability of the measurement rather than whether the watch is being worn.

The Labfront dashboard provides additional indicators including No HR Data, No BBI Data, and Haven't Synced, which can help identify periods where physiological data are unavailable. However, these indicators do not distinguish between a participant removing the watch, poor sensor contact, signal quality issues, or the device simply not synchronizing with Garmin.



Labfront Dashboard

From what I see, non-wear cannot be directly identified from the exported data. Instead, it must be inferred through missing physiological measurements, synchronization status, or prolonged periods of low-quality signals.

Sync Delay

According to the Labfront documentation, physiological data collected through the Labfront app typically takes 2–5 minutes to become available after synchronization, with BBI uploads sometimes taking longer due to the higher volume of data.

For the current decision engine, this delay is unlikely to significantly affect EMA-based interventions, though it should be considered if future versions require immediate physiological responses.